

📦 Project Transfer Guide — DeepAgent to Mac Mini M4 Pro

Transfer the Final Evolution Lab project from DeepAgent to your local Mac Mini M4 Pro

Project Size: ~500 MB – 2 GB (depending on generated assets)

Estimated Time: 5-30 minutes (depending on method and connection speed)

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Option A: Download as ZIP from DeepAgent

★ **Easiest method — no command line needed**

🕒 **Time: 5-15 minutes**

Step 1: Download from DeepAgent UI

1. In the DeepAgent interface, locate the project files
2. Click the **“Download”** or **“Export”** button in the Code Editor UI
3. Select **“Download as ZIP”**
4. Wait for the ZIP to download to your Mac’s ~/Downloads/ folder

Step 2: Extract on Your Mac

Create project directory

```
mkdir -p ~/Projects
```

Extract the ZIP

```
cd ~/Projects
```

```
unzip ~/Downloads/rork-final-evolution-lab.zip
```

```
# Or if the ZIP has a different name
unzip ~/Downloads/rork-final-evolution-lab*.zip -d ~/Projects/
```

Step 3: Verify

```
# Check project structure
ls ~/Projects/rork-final-evolution-lab/
# Should see: UnrealStarter/ scripts/ streaming/ deployment/
# etc.

# Check key files exist
test -f ~/Projects/rork-final-evolution-
lab/UnrealStarter/BasketballGame/BasketballGame.uproject &&
echo "✓ UE5 project file found" || echo "✗ Missing!"
test -d ~/Projects/rork-final-evolution-lab/deployment/aws && echo
"✓ Terraform configs found" || echo "✗ Missing!"
test -d ~/Projects/rork-final-evolution-lab/streaming && echo "✓
Streaming infrastructure found" || echo "✗ Missing!"
```

Option B: Git Clone

☆ **Recommended** — keeps version history and enables easy updates

🕒 **Time: 5-10 minutes**

Prerequisites

- Git installed on your Mac (comes with Xcode Command Line Tools)
- GitHub account with access to the repository

Step 1: Install Git (if needed)

```
# Check if Git is installed
git --version

# If not, install via Xcode Command Line Tools
xcode-select --install

# Or via Homebrew
brew install git
```

Step 2: Configure Git

```
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"
```

Step 3: Clone the Repository

```
mkdir -p ~/Projects
cd ~/Projects

# Clone via HTTPS
git clone https://github.com/YOUR_USERNAME/rork-final-evolution-
lab.git

# OR clone via SSH (if you have SSH keys set up)
git clone git@github.com:YOUR_USERNAME/rork-final-evolution-lab.git
```

Replace YOUR_USERNAME with the actual GitHub username/org.

Step 4: Set Up Environment

```
cd ~/Projects/rork-final-evolution-lab

# Copy environment template
cp .env.example .env

# Edit with your API keys
nano .env
```

Step 5: Verify

```
git status
git log --oneline -5
```

Option C: SCP/Rsync Transfer

🔗 **Advanced** — direct transfer if you have SSH access to DeepAgent VM

🕒 **Time: 10-30 minutes**

This method requires SSH access to the DeepAgent VM. The PEM key files in your Uploads folder may be used for this.

Step 1: Locate Your PEM Key

```
# Check for the PEM key on your Mac
ls ~/Downloads/*.pem

# Or wherever you saved the uploaded PEM files

# Set permissions
chmod 600 ~/Downloads/FinalEvolutionLab.pem
```

Step 2: SCP Transfer (Simple)

```
mkdir -p ~/Projects
```

```
# Transfer entire project
```

```
scp -i ~/Downloads/FinalEvolutionLab.pem -r \  
    ubuntu@DEEPAGENT_IP:/home/ubuntu/rork-final-evolution-lab \  
    ~/Projects/
```

Replace DEEPAGENT_IP with the DeepAgent VM's IP address.

Step 3: Rsync Transfer (Better for Large Projects)

```
# Rsync with compression and progress (resumes interrupted transfers)
```

```
rsync -avz --progress \  
    -e "ssh -i ~/Downloads/FinalEvolutionLab.pem" \  
    --exclude '.git' \  
    --exclude 'node_modules' \  
    --exclude 'Intermediate' \  
    --exclude 'Saved' \  
    --exclude '.terraform' \  
    ubuntu@DEEPAGENT_IP:/home/ubuntu/rork-final-evolution-lab/ \  
    ~/Projects/rork-final-evolution-lab/
```

Step 4: Transfer Generated Assets Separately (if large)

```
# Transfer just the GeneratedAssets folder
```

```
rsync -avz --progress \  
    -e "ssh -i ~/Downloads/FinalEvolutionLab.pem" \  
    ubuntu@DEEPAGENT_IP:/home/ubuntu/rork-final-evolution-  
        lab/GeneratedAssets/ \  
    ~/Projects/rork-final-evolution-lab/GeneratedAssets/
```

Option D: S3 Intermediate Transfer

🔧 **Advanced** — useful if direct SCP is slow or unreliable

🕒 **Time: 15-30 minutes**

On DeepAgent (or ask DeepAgent to run these):

```
# Install AWS CLI on DeepAgent (if not installed)
```

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o  
    "awscliv2.zip"
```

```
unzip awscliv2.zip && sudo ./aws/install
```

```
# Configure AWS credentials
```

```
aws configure
```

```
# Create a tarball of the project
cd /home/ubuntu
tar -czf rork-final-evolution-lab.tar.gz \
  --exclude='rork-final-evolution-lab/.git' \
  --exclude='rork-final-evolution-lab/node_modules' \
  --exclude='rork-final-evolution-lab/.terraform' \
  rork-final-evolution-lab/

# Upload to S3
aws s3 cp rork-final-evolution-lab.tar.gz
      s3://YOUR_BUCKET/transfer/rork-final-evolution-lab.tar.gz
```

On Your Mac:

```
mkdir -p ~/Projects
cd ~/Projects

# Download from S3
aws s3 cp s3://YOUR_BUCKET/transfer/rork-final-evolution-lab.tar.gz
      .

# Extract
tar -xzf rork-final-evolution-lab.tar.gz

# Clean up
rm rork-final-evolution-lab.tar.gz

# Clean up S3 (to avoid storage costs)
aws s3 rm s3://YOUR_BUCKET/transfer/rork-final-evolution-lab.tar.gz
```

Post-Transfer Verification

Run these checks after any transfer method:

```
cd ~/Projects/rork-final-evolution-lab

echo "=== Project Structure Check ==="
for dir in UnrealStarter streaming deployment scripts ios
  GeneratedAssets; do
  if [ -d "$dir" ]; then
    echo "✔ $dir/ exists"
  else
    echo "✗ $dir/ MISSING"
  fi
done

echo ""
echo "=== Key Files Check ==="
```

```

for file in \
    UnrealStarter/BasketballGame/BasketballGame.uproject \
    deployment/aws/main.tf \
    streaming/signalling/server.js \
    streaming/frontend/package.json \
    scripts/ue5_setup/cook_fel_linux_server.sh \
    DEPLOYMENT_CHECKLIST.md \
    MAC_MINI_SETUP_GUIDE.md \
    MAC_AWS_DEPLOYMENT.md; do
    if [ -f "$file" ]; then
        echo "✔ $file"
    else
        echo "✗ $file MISSING"
    fi
done

echo ""
echo "=== Project Size ==="
du -sh .

echo ""
echo "=== Generated Assets ==="
find GeneratedAssets -type f | wc -l | xargs -I{} echo "{} files in
GeneratedAssets/"

```

Expected Output

```

=== Project Structure Check ===
✔ UnrealStarter/ exists
✔ streaming/ exists
✔ deployment/ exists
✔ scripts/ exists
✔ ios/ exists
✔ GeneratedAssets/ exists

=== Key Files Check ===
✔ UnrealStarter/BasketballGame/BasketballGame.uproject
✔ deployment/aws/main.tf
✔ streaming/signalling/server.js
✔ streaming/frontend/package.json
✔ scripts/ue5_setup/cook_fel_linux_server.sh
✔ DEPLOYMENT_CHECKLIST.md
✔ MAC_MINI_SETUP_GUIDE.md
✔ MAC_AWS_DEPLOYMENT.md

=== Project Size ===
~500 MB - 2 GB

```

```
=== Generated Assets ===  
XX files in GeneratedAssets/
```

Troubleshooting

ZIP Download is Too Large

If the project exceeds DeepAgent's download limit:

```
# Ask DeepAgent to create a smaller archive excluding large files  
# Then transfer GeneratedAssets separately via S3
```

Git Clone Fails — “Repository Not Found”

- Verify the repository URL is correct
- Check you have access (try opening the URL in a browser)
- Use HTTPS if SSH isn't configured:

```
git clone https://github.com/YOUR_USERNAME/rork-final-evolution-  
lab.git
```

SCP Connection Refused

- The DeepAgent VM may have shut down (VMs are temporary)
- Use Git clone or S3 transfer instead
- Check that the PEM file permissions are correct: `chmod 600 *.pem`

Missing Files After Transfer

- Check `.gitignore` — some files may not be tracked:
 - `.env` (contains API keys — recreate from `.env.example`)
 - `node_modules/` (reinstall with `npm install`)
 - `.terraform/` (reinitialize with `terraform init`)
 - `Builds/` (rebuild with UE5)

```
# Reinstall Node dependencies  
cd streaming/signalling && npm install  
cd ../frontend && npm install
```

```
# Copy and configure .env  
cp .env.example .env  
nano .env
```

```
# Re-initialize Terraform  
cd deployment/aws && terraform init
```

Transfer Speed is Slow

Use compression with rsync

```
rsync -avz --compress-level=9 ...
```

Or split into smaller transfers

```
rsync -avz --include='UnrealStarter/**' --exclude='*' ...
```

```
rsync -avz --include='streaming/**' --exclude='*' ...
```

```
rsync -avz --include='deployment/**' --exclude='*' ...
```

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